BHUMIverse Project Proposal

Intelligent Land Record Verification & Fraud Prevention Platform

LandTech Innovation Hackathon 2026

Team name: BHUMIverse | Primary category: P3 Catch Fraud & Double-Selling

Track: Digital Land Records, Verification and Digitization | Date: 18 October 2026

1 Project Summary

BHUMIverse is an intelligent land fraud detection and double-selling prevention platform engineered specifically for Bangladesh. Our platform solves Hackathon category P3 ('Catch Fraud & Double-Selling') by identifying fake and tampered documents, catching cases where the same land is sold to more than one buyer, and revealing dangerous record mismatches between deed registration (Sub-Registry) and mutation (AC Land offices). Citizens and buyers can search by parcel or upload deeds to inspect altered text, verify seller decimal limits, and uncover organized fraud syndicates. The platform delivers an explainable Land Verification Report with an instant Risk Score (Low, Medium, High) and clear next steps in both English and Bengali.

2 Selected Problem and Objectives

Primary Category: P3 Catch Fraud & Double-Selling (Spot fake documents, double-selling, and registration-mutation mismatches).

In Bangladesh, over 70% of all civil disputes arise from fraudulent land transactions. Citizens face four severe fraud threats: (1) Fake and tampered deeds where photocopy text, dates, or plot sizes are digitally altered; (2) Double-selling where dishonest sellers sell the same plot to multiple buyers before any mutation is recorded; (3) Registration-mutation mismatch where registered deeds never get mutated at the AC Land office, leaving the seller's name in official records to enable second illegal sales; and (4) Fraud syndicates that exploit loopholes across disjointed government offices.

Core Objectives: (a) Spot fake and altered documents using automated Smart Document Diffing; (b) Mathematically detect double-selling and over-selling by tracking seller remaining decimals; (c) Reconcile deed registration records with mutation khatians to flag broken ownership chains; and (d) Map repeat fraud actors and disputed plots on an interactive fraud network graph.

3 Existing Services and Main Contribution

Current government services like e-Porcha, e-Mutation, and online Land Tax operate in silos. Sub-Registry offices record deeds without real-time verification of the seller's live remaining balance at the AC Land office. This gap allows dishonest sellers to sell the same plot multiple times. BHUMIverse connects these separated records into an explained fraud-detection matrix.

Existing Government Baseline	Limitations in Catching Fraud	BHUMIverse Platform Contribution
e-Porcha & Record Lookup (land.gov.bd)	Shows individual survey records but cannot verify whether the seller has already sold their share to someone else.	Builds an automated Transaction Timeline connecting CS-SA-RS-BS and tracking live remaining seller decimals.
Sub-Registry Deed Registration	Registers deeds based on presented papers without verifying live mutation records or checking duplicate sales.	Deterministic Rule Engine cross-checks past deeds, catches duplicate sales, and identifies registered deeds lacking mutation.
Photocopy Document Verification	Officers and buyers have no automated tool to detect altered	Smart Document Diff tool provides side-by-side visual and OCR comparison to highlight tampered text and

	numbers, dates, or names on photocopy deeds.	numbers in red.
Fraud Syndicate Monitoring	No government system connects repeat fraudsters across different sub-registries and AC land jurisdictions.	Fraud Network Graph visually models buyers, sellers, brokers, and disputed plots on an interactive node-link graph.

4 System Workflow

The BHUMIverse platform follows a structured 7-step fraud detection workflow:

- **Step 1 (Access Platform):** Citizen accesses the web portal and logs in (or tests pre-loaded benchmark fraud cases).
- **Step 2 (Select Land Parcel):** User enters District, Upazila, Mouza, JL Number, Dag, and Khatian number to retrieve historical records.
- **Step 3 (Upload Evidence):** User uploads scanned deeds or porchas. The system generates a cryptographic SHA-256 fingerprint to immediately flag duplicate file submissions.
- **Step 4 (Run Fraud Engine):** User enters proposed purchase area. The engine runs 5 deterministic fraud rules: remaining seller share, survey continuity, registered mortgage, court litigation, and khas/acquisition status.
- **Step 5 (Visual Diagnostics):** Results are displayed on an Explainable Radar Chart (covering Authenticity, Continuity, Custody, Encumbrance, and Consistency) and an interactive Fraud Network Graph.
- **Step 6 (Download Audit Report):** Platform generates a comprehensive Land Verification Report with an overall Risk Score (Low, Medium, High), audit verification hash, and downloadable PDF report.
- **Step 7 (Case Review & Next Steps):** Citizens or reviewers can flag contested records for secondary human review and log immutable audit events.

Real-World Double-Selling Example: A seller owns 10 decimal in Savar Mouza (Dag 10482). In 2021, the seller sold 6 decimal to Buyer A via Deed No. 4821. Buyer A delayed filing for mutation. In 2023, the seller produces an old khatian copy showing 10 decimal and attempts to sell 8 decimal to Buyer B. The BHUMIverse rule engine analyzes past deed transfers and discovers that only 4 decimal remain. When 8 decimal is entered, the system triggers a Critical Double-Selling Alert ($6 + 8 = 14 > 10$ decimal), stopping the fraudulent sale immediately.

5 Architecture and Implementation Method

BHUMIverse is engineered with modern, robust, scalable technologies:

- **Frontend Web Application:** Next.js 16 (App Router), React 19, TypeScript, and Tailwind CSS. Features Vis-network for interactive fraud syndicate graphs, Chart.js/SVG for explainable radar charts, and full dual-language support (English and Bengali).
- **Backend REST API:** FastAPI (Python 3.11) with Pydantic v2 schemas. Implements deterministic rule engines, SHA-256 file fingerprinting, and audit logging services.
- **Database & Authentication:** Supabase (PostgreSQL with Row-Level Security) managing user accounts, parcel metadata, case histories, and audit trails.
- **Smart Document Diff Engine:** Side-by-side visual difference tool that compares original deed templates against altered uploaded documents to highlight modified text or numbers.
- **Deployment:** Vercel production build for frontend and cloud-ready Dockerized backend architecture.

6 Actual Development Progress

Status as of 18 October 2026: The BHUMIverse prototype is fully built, tested, and operational. All core features can be tested live.

P3 Fraud Module	Development Status	Demonstrated Capabilities in Prototype

Double-Selling Detection Engine	100% Implemented	Computes remaining seller share balance; alerts when proposed sale exceeds owned decimals.
Smart Document Diff Tool	100% Implemented	Side-by-side deed diffing; visual red-highlighting of altered text and dates; confidence scoring.
Fraud Network Graph	100% Implemented	Interactive physics-based node graph showing linked buyers, sellers, disputed plots, and fraud syndicates.
Registration-Mutation Mismatch	100% Implemented	Cross-checks registered deeds against AC Land mutation khatians to flag non-mutated ghost titles.
Explainable Radar Chart	100% Implemented	5-axis radar chart showing exact scores across Authenticity, Continuity, Custody, Encumbrance, and Consistency.
Transaction Timeline	100% Implemented	Chronological card timeline tracking ownership transfers across CS, SA, RS, and BS survey records.
Bilingual Support (EN / BN)	100% Implemented	Instant one-click toggle between English and Bengali across all pages, forms, and results.
Supabase Auth & Audit Logs	100% Implemented	Secure sign-in, case creation, document uploads, and immutable audit logs.

7 Legal Alignment and Data Protection

BHUMIverse is designed in strict compliance with the statutory land laws of Bangladesh:

- **Transfer of Property Act 1882 (Sections 53C & 53D):** Under Section 53C, any land transfer is legally void if the seller's name is not recorded in the latest khatian (or supported by a valid succession certificate). Under Section 53D, transferring mortgaged property without the lender bank's written consent is illegal. BHUMIverse enforces both checks automatically.
- **Registration Act 1908 (Section 52A & 2026 Amendment):** Mandates a complete 25-year chain of title and verified mutation porcha before any deed registration. The BHUMIverse timeline directly verifies this 25-year continuity.
- **Penal Code 1860 (Sections 420 & 468 - Fraud & Forgery):** Categorizes deed forgery and cheating as criminal offenses. BHUMIverse identifies tampered photocopies and fraudulent double-selling patterns to protect citizens from criminal deception.
- **Data Privacy & Ethics:** All demonstration data is synthetic and non-confidential. Citizen records are protected with encryption and Row-Level Security (RLS). No unauthorized scraping of government portals is conducted.

8 Limitations and Mitigation

Current Prototype Limitation	Mitigation Strategy & Future Solution
No public API available for live government land databases during hackathon.	Developed standardized JSON adapters and a realistic 8-parcel synthetic benchmark covering real-world Savar, Dhanmondi, and Mirpur scenarios.
Handwritten historical deeds in archaic Bangla script are difficult for raw OCR.	Integrated human-in-the-loop confirmation where the system extracts proposed values and prompts the user to confirm before running rules.
Automated software cannot replace an official legal court decree or title certificate.	BHUMIverse explicitly presents itself as an advisory verification tool with clear checklists, advising citizens when to hire a qualified land lawyer.

9 Future Scope in Bangladesh

- **1. Upazila-Level Pilot:** Launch a controlled pilot in one Upazila (e.g., Savar or Gazipur Sadar) in collaboration with the local AC Land office and Sub-Registry.

- **2. Official API Integrations:** Connect directly with the Land Ministry's e-Mutation API, National ID (NID/Porichoy) verification, and the National Land Zoning database.
- **3. Citizen Accessibility:** Deploy offline-capable kiosks at Union Digital Centers (UDCs) and launch an Android mobile app with Bangla voice guidance for rural farmers and overseas expatriates.

10 Evaluation and Delivery Plan

For Round 1 of the LandTech Innovation Hackathon 2026, the BHUMIverse team delivers:

- **1. Project Proposal:** Complete 4-page concept note and system architecture document focused on P3 fraud prevention.
- **2. Working Prototype:** Live, interactive web platform featuring document diffing, fraud network visualization, rule-based verification, and instant report generation.
- **3. Source Code Repository:** Fully documented open-source repository with setup guides, test suites, and synthetic datasets.
- **4. Video Demonstration:** Comprehensive video walkthrough demonstrating all citizen verification flows within 3 minutes.

11 Team and Mentor Information

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12 References and Prototype

- **[1] DIU CSE and CSE-TECH:** LandTech 2026 Rule Book v1.2, Department of CSE & CSE-TECH, Daffodil International Universit
- **[2] Ministry of Land, Bangladesh:** e-Mutation, e-Porcha, and Land Development Tax Portals (land.gov.bd).
- **[3] Transfer of Property Act 1882:** Sections 53C & 53D (Transfer restrictions, khatian recording, mortgage consent).
- **[4] Registration Act 1908 & 2026 Amendment:** Section 52A (Mandatory 25-year title chain check and deed format).
- **[5] Penal Code 1860:** Sections 420 & 468 (Cheating and document forgery offenses).
- **[6] BHUMIverse Open-Source Repository:** https://github.com/parvezrepository/bhumiverse